

Solutions to JEE Advanced – 4 | JEE 2022 | Paper-1

PHYSICS

1.(A) Field due to a side of length a

$$B_1 = \frac{\mu_0 I}{2\pi \left(\frac{b}{2}\right)} \left(\left(\frac{a}{\sqrt{a^2 + b^2}} \right) \right) = \frac{\mu_0 I}{\pi} \left(\frac{a}{b\sqrt{a^2 + b^2}} \right)$$

Similarly, the field due to a side of length b

$$B_2 = \frac{\mu_0 I}{\pi} \left(\frac{b}{a\sqrt{a^2 + b^2}} \right)$$

We can check by using right hand rule that the field due to all the sides will be in the same direction (all perpendicular to the plane of the loop and in the same direction), and hence we can simply add the contributions of the sides

Hence, net field at the centre, $B_{NET} = 2B_1 + 2B_2 = \frac{2\mu_0 I}{\pi} \left(\frac{\sqrt{a^2 + b^2}}{ab} \right)$

2.(B) Resistance of the shell, $r = \int_R^{3R} \left(\frac{\rho}{2\pi x L} \right) dx = \left(\frac{\rho}{2\pi L} \right) \log_e(3)$

So, the current through the shell, $I = \frac{V}{r} = \frac{2\pi L V}{\rho \log_e(3)}$

Electric field at a point at a distance $2R$ from the axis of the shell is $E = \rho j = \rho \left(\frac{i}{2\pi(2R)L} \right) = \frac{V}{2R \log_e(3)}$

3.(B) Let the velocity of the particle when it enters the magnetic field be v

We know that

Change in kinetic energy of the particle = Work done on the particle by the electric field

Therefore, $\frac{1}{2}mv^2 = qEd \Rightarrow v = \sqrt{\frac{2qEd}{m}}$

Now, the radius of the circular path of the particle in the magnetic field, $r = \frac{mv}{qB} = \frac{1}{B} \sqrt{\frac{2mEd}{q}}$

So, $y_0 = 2r = \frac{2}{B} \sqrt{\frac{2mEd}{q}} = \left(\frac{8mEd}{qB^2} \right)^{1/2}$

4.(B) Let the area of the loop be A , and let its angular velocity be ω

Let us define a vector $\hat{n}$ perpendicular to the plane of the loop.

Let the instantaneous angle between the vector $\hat{n}$ and the magnetic field be θ

Then, magnetic flux through the loop is $\phi = B(A \cos \theta)$

But, $\theta = \omega t$ $\phi = B(A \cos \omega t)$

Therefore, instantaneous induced current in the loop, $i = \left| \frac{1}{R} \frac{d\phi}{dt} \right| = \left| \frac{BA\omega}{R} \sin \omega t \right|$

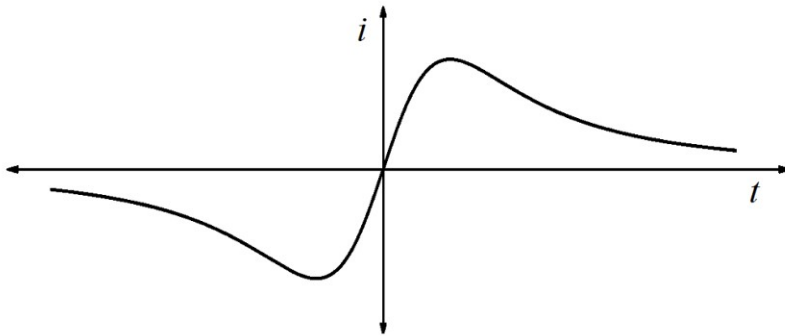
So, maximum value of this induced current, $i_{MAX} = \frac{BA\omega}{R} = \frac{B(L^2)\omega}{\beta(4L)} = \frac{BL\omega}{4\beta}$

5.(A) We know that $I = +\frac{dQ}{dt} \Rightarrow \int_0^Q dQ = \int_0^t (I_0 \cos(\omega t)) dt \Rightarrow Q(t) = \frac{I_0}{\omega} \sin(\omega t)$

Remember that in a series LCR (or RC) circuit, the charge on the capacitor is always behind the current in the circuit by a phase angle $\frac{\pi}{2}$. This does not depend on the values of R , L or C or the frequency.

Hence the given angular frequency $\omega = \frac{1}{RC}$ is of no consequence to the question asked.

- 6.(C) For the first half of the journey of the magnet, the flux through the solenoid is increasing, and for the second half, the flux is decreasing. Hence, the current will reverse direction exactly once.
Therefore, the induced current is zero at the instant the magnet is at the centre of the solenoid
Also, initially the magnet is at a large distance from the solenoid, and so initially, the flux through the solenoid changes very slowly, and hence the induced current is initially zero
By the same reasoning, the induced current becomes zero finally as well
So, the variation of the induced current is as follows:



7.(AD) Area of the loop, $A = \pi a^2$
Resistance of the loop, $R = \frac{\rho(2\pi a)}{\left(\frac{\pi d^2}{4}\right)} = \frac{8\rho a}{d^2}$

Instantaneous induced current, $I = \frac{A}{R} \frac{dB}{dt} = \frac{\pi B_0 \omega a d^2}{8\rho} \cos(\omega t)$

Therefore, maximum current, $I_M = \frac{\pi B_0 \omega a d^2}{8\rho}$

Instantaneous rate of heat dissipation, $H = I^2 R = \left(\frac{\pi B_0 \omega a d^2}{8\rho} \cos(\omega t) \right)^2 \left(\frac{8\rho a}{d^2} \right) = \frac{\pi^2 B_0^2 \omega^2 a^3 d^2}{8\rho} \cos^2(\omega t)$

Since the average value of $\cos^2(\omega t)$ over many cycles is $1/2$, $H_{AV} = \frac{\pi^2 B_0^2 \omega^2 a^3 d^2}{16\rho}$

- 8.(ABD) (A) We can check this by drawing a diagram of the X-Z plane
(B) We can see that the electric field at all points of the Y-Z plane is in the $-X$ direction, and hence perpendicular to the Y-Z plane. Therefore, the Y-Z plane is an equipotential surface
(C) At $(x, 0, 0)$, $\vec{E} = \left[\frac{\lambda}{2\pi\epsilon_0(a-x)} \right] (-\hat{i}) + \left[\frac{\lambda}{2\pi\epsilon_0(a+x)} \right] (-\hat{i}) = \left[\frac{\lambda a}{\pi\epsilon_0(a^2 - x^2)} \right] (-\hat{i})$

$$\begin{aligned}
 \text{(D) Work done, } W &= -(q) \int_{x=0}^{x=\frac{a}{2}} \vec{E} \cdot d\vec{x} = \int_0^{\frac{a}{2}} \frac{\lambda q a}{\pi \epsilon_0 (a^2 - x^2)} dx = \frac{\lambda q}{2\pi \epsilon_0} \left[\int_0^{\frac{a}{2}} \frac{dx}{a-x} + \int_0^{\frac{a}{2}} \frac{dx}{a+x} \right] \\
 &= \frac{\lambda q}{2\pi \epsilon_0} \log_e \left| \frac{a+x}{a-x} \right|_0^{\frac{a}{2}} = \frac{\lambda q}{2\pi \epsilon_0} \log_e 3
 \end{aligned}$$

9.(ABD) Current leads the voltage by a phase difference:

$$\phi = \tan^{-1} \left(\frac{\frac{1}{\omega C} - \omega L}{R} \right) = \tan^{-1} \left(\frac{\frac{1}{(100)(10 \times 10^{-6})} - (100)(1)}{R} \right) = 45^\circ \Rightarrow R = 900 \Omega$$

$$\text{(A) } P_{AV} = I_{RMS}^2 R \Rightarrow I_{RMS} = \sqrt{\frac{P_{AV}}{R}} = \frac{1}{\sqrt{10}} A$$

$$\text{(B) Power factor } \cos \phi = \frac{R}{Z} = \frac{450}{\sqrt{450^2 + 450^2}} = \frac{1}{\sqrt{2}}$$

$$\text{(C) } V_M = \sqrt{2} I_{RMS} Z = (\sqrt{2}) \left(\frac{1}{\sqrt{10}} \right) (900\sqrt{2}) = 180\sqrt{10} \text{ V}$$

$$\text{(D) Current lags behind the voltage source by } \frac{\pi}{4} \Rightarrow \frac{\omega L - \frac{1}{\omega C}}{R} = \tan 45^\circ = 1 \Rightarrow \omega = 1000 \text{ rad/s}$$

10.(ACD) Resistance of the portions PS and SQ is

$$R_{PS} = \left(\frac{PS}{PQ} \right) (40) = 30 \Omega \quad \text{and} \quad R_{SQ} = \left(\frac{SQ}{PQ} \right) (40) = 10 \Omega$$

$$\text{Potential difference between P and S is } V_P - V_S = (i_1 - i_2) R_{PS} = (0.8) \left(\frac{45}{2} \right) = 24 \text{ V}$$

Since it is ideal, the EMF of the battery E_1 must be equal to V_{PQ}

$$\text{So, } E_1 = V_{PQ} = V_P - V_Q = (i_1 - i_2) R_{PS} + i_1 R_{SQ} = (0.8)(30) + (1.8)(10) = 42 \text{ V}$$

Starting from P and applying KVL in the small loop, going clockwise:

$$-(i_1 - i_2)(30) + E_2 + i_2 r_2 = 0 \Rightarrow -(0.8)(30) + 20 + r_2 = 0 \Rightarrow r_2 = 4 \Omega$$

Since $E_2 < E_1$, current through the branch containing E_2 can be made zero by changing length PS.

11.(AB) We know that the current in the top branch varies as

$$I_L(t) = \frac{E}{R} (1 - e^{-Rt/L}) = 2(1 - e^{-K_1 t}) \quad \text{where } K_1 = 10000 \text{ s}^{-1}$$

And we know that the potential difference across the capacitor varies as

$$V_C(t) = E(1 - e^{-t/RC}) = 100(1 - e^{-K_2 t}) \quad \text{where } K_2 = 5000 \text{ s}^{-1}$$

$$\text{Using KVL, the reading of the voltmeter varies as } V(t) = -V_C(t) + I_L(t)R = 100(e^{-K_2 t} - e^{-K_1 t})$$

Putting the derivative of this reading to zero to find the maximum,

$$\frac{dV(t)}{dt} = 100(-K_2 e^{-K_2 t} + K_1 e^{-K_1 t}) = 0 \Rightarrow t = \left(\frac{1}{K_1 - K_2} \right) \log_e \left(\frac{K_1}{K_2} \right) = (2 \log_e 2) \times 10^{-4} \text{ s}$$

This is the time instant when $V(t)$ reaches its maximum value. This maximum value is

$$V_{MAX} = 100(2^{-1} - 2^{-2}) = 25 \text{ V}$$

12.(BCD) Field at S due to side PQ is in the $+Z$ direction.

The perpendicular distance of S from PQ , $r_{PQ} = \frac{a\sqrt{3}}{2}$

So, field due to PQ , $B_{PQ} = \frac{\mu_0 I}{4\pi r_{PQ}} (\cos 30^\circ + \cos 120^\circ) = \frac{\mu_0 I}{4\pi a} \left(1 - \frac{1}{\sqrt{3}}\right)$

Field at S due to side RP is also in the $+Z$ direction and has the same magnitude due to symmetry.

Now, field at S due to side QR is in the $-Z$ direction.

The perpendicular distance of S from QR ,

$$r_{QR} = \frac{a\sqrt{3}}{2}$$

So, field due to QR , $B_{QR} = \frac{\mu_0 I}{4\pi r_{QR}} (\cos 60^\circ + \cos 60^\circ) = \frac{\mu_0 I}{2\sqrt{3}\pi a}$

Net field at S , $\vec{B}_{net} = \vec{B}_{PQ} + \vec{B}_{QR} + \vec{B}_{RP} = \left[\frac{\mu_0 I}{2\pi a} \left(\frac{2}{\sqrt{3}} - 1 \right) \right] (-\hat{k})$

13.(1.57)

Due to the rotation of the loop, a current is induced in it, and due to this induced current, the magnetic field applies a torque on the loop. We can check easily that this torque will oppose the rotation of the loop

At an instant when the loop has rotated by angle θ from its initial position, the flux through it is

$$\Phi = BL^2 \cos \theta$$

Therefore, at this instant the induced current is $i = \frac{1}{R} \left| \frac{d\Phi}{dt} \right| = \frac{BL^2}{R} \sin \theta \left(\frac{d\theta}{dt} \right) = \frac{BL^2 \omega}{R} \sin \theta$

Here, ω is the instantaneous angular velocity of the loop

So, the instantaneous magnetic moment of the loop is $M = iL^2 = \frac{BL^4 \omega}{R} \sin \theta$

And, the instantaneous torque on it is $\tau = MB \sin \theta = \frac{B^2 L^4 \omega}{R} \sin^2 \theta$

Let the moment of inertia of the loop about the Y-axis be I_Y

Then, if the angular velocity of the loop immediately after it is given the angular impulse is ω_0 , $J = I_Y \omega_0$

Also, since we know that the magnetic torque on the loop opposes its angular motion,

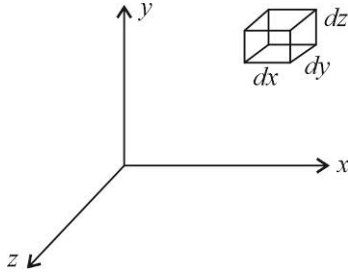
$$\tau = -I_Y \left(\omega \frac{d\omega}{d\theta} \right) \quad \left(\omega \frac{d\omega}{d\theta} \right) \text{ is the angular acceleration of the loop}$$

$$\Rightarrow I_Y \left(\omega \frac{d\omega}{d\theta} \right) = -\frac{B^2 L^4 \omega}{R} \sin^2 \theta$$

$$\Rightarrow I_Y \int_{\omega_0}^0 d\omega = -\frac{B^2 L^4}{R} \int_0^\pi \sin^2 \theta d\theta$$

$$\Rightarrow I_Y \omega_0 = \frac{B^2 L^4}{R} \left(\frac{\pi}{2} \right) \Rightarrow J = \left(\frac{\pi}{2} \right) \frac{B^2 L^4}{R}$$

14.(4)

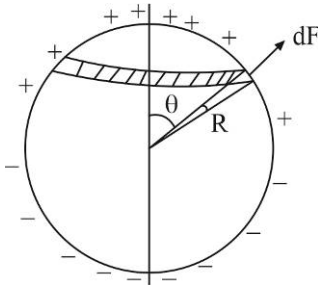


$$\phi = (E_{(x+dx)} - E_x) dydz + (E_{(y+dy)} - E_y) dx dz + (E_{(z+dz)} - E_z) dx dy = \frac{\rho(dx dy dz)}{\epsilon_0}$$

$$\Rightarrow \frac{\rho}{\epsilon_0} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \Rightarrow \rho = \epsilon_0 (2x - 2y + 2) = 4\epsilon_0$$

15.(0.08) Motional EMF induced across the ends of the rod, $\epsilon = \frac{1}{2} B \omega L^2 = \frac{1}{2} (2)(8)(0.1)^2 = 0.08 \text{ V}$

16.(0.25)



$$dF = \frac{\sigma E dA}{2}; E = \frac{\sigma}{\epsilon_0}$$

$$dA = (2\pi R \sin \theta) R d\theta$$

$$F_{net} = \int dF \cos \theta = \int_0^{\pi/2} \frac{\pi \sigma^2 R^2}{\epsilon_0} \sin \theta \cos \theta d\theta = \frac{\pi \sigma_0^2 R^2}{4 \epsilon_0}$$

17.(2) For the first capacitor, $\frac{1}{C_{eq}} = \frac{\left(\frac{d}{3}\right)}{K\epsilon_0 A} + \frac{\left(\frac{2d}{3}\right)}{4K\epsilon_0 A} = \frac{d}{2K\epsilon_0 A} \Rightarrow C_{eq} = \frac{2K\epsilon_0 A}{d}$

For the second capacitor, $C_{eq} = \frac{K_0 \epsilon_0 A}{d} \Rightarrow \frac{K_0}{K} = 2$

18.(11.25) Potential energy of a dipole kept in a uniform field, $U = -\vec{p} \cdot \vec{E} = -pE \cos \theta$, where θ is the angle between the dipole moment vector and the electric field vector.

Work done in rotating a dipole kept in a uniform electric field is

$$U_f - U_i = (-pE(-1)) - (-pE(1)) = 2pE = 2(p) \left(\frac{\sigma}{2\epsilon_0} \right) = 11.25 \text{ J}$$

CHEMISTRY

$$1.(C) \quad Y_A = \frac{P_A^\circ X_A}{P_A^\circ X_A + P_B^\circ X_B}$$

$$\frac{1}{Y_A} = 1 + \frac{P_B^\circ}{P_A^\circ} \frac{X_B}{X_A}$$

$$\frac{1}{Y_A} = 1 + \frac{P_B^\circ}{P_A^\circ} \frac{(1 - X_A)}{X_A} = \left(1 - \frac{P_B^\circ}{P_A^\circ}\right) + \frac{P_B^\circ}{P_A^\circ} \cdot \frac{1}{X_A}$$

$$\left(\frac{1}{Y_A} - \left(1 - \frac{P_B^\circ}{P_A^\circ}\right)\right) \frac{P_A^\circ}{P_B^\circ} = \frac{1}{X_A}$$

$$\frac{1}{X_A} = \frac{P_A^\circ}{P_B^\circ} \cdot \frac{1}{Y_A} + \left(\frac{P_B^\circ - P_A^\circ}{P_B^\circ}\right)$$

$$2.(D) \quad \frac{P_A^\circ - P_S}{P_S} = \frac{n}{N}$$

$$\frac{89.78 - 89}{89} = \frac{(2 / M_o)}{(100 / 78)} \Rightarrow M_o = 178 \text{ g / mol}$$

$$\%C \text{ by mass of } C_xH_y \text{ hydrocarbon} = \frac{12x}{178} \times 100 = 94.4$$

$$x = 14 \quad \text{Formula of hydrocarbon} = C_{14}H_{10}$$

$$3.(D) \quad 2.5 \times 10^{-5} = \frac{1}{100 \times 60} \ln \frac{[SO_2Cl_2]_o}{[SO_2Cl_2]_t}$$

$$0.15 = \ln \frac{[SO_2Cl_2]_o}{[SO_2Cl_2]_t} \Rightarrow \frac{[SO_2Cl_2]_o}{[SO_2Cl_2]_t} = 1.16$$

$$\% SO_2Cl_2 \text{ decomposed} = \frac{[SO_2Cl_2]_o - [SO_2Cl_2]_t}{[SO_2Cl_2]_o} \times 100 = \left(1 - \frac{1}{1.16}\right) \times 100 = 13.8\%$$

$$4.(C) \quad \frac{x}{m} = KP^n$$

$$0.2 = K \times (4)^{\frac{1}{n}} \dots (1)$$

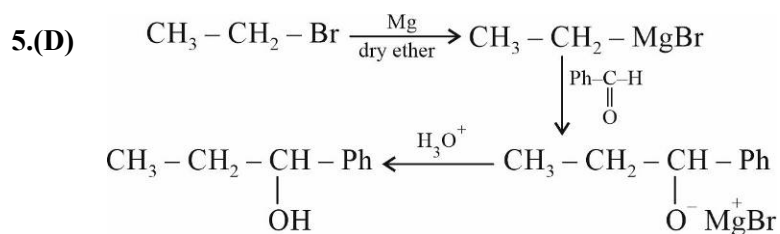
$$0.5 = K \times (25)^{\frac{1}{n}} \dots (2)$$

$$\frac{2}{5} = \left(\frac{4}{25}\right)^{\frac{1}{n}} \Rightarrow n = 2, K = 0.1$$

$$\frac{x}{m} \text{ at 36 bar pressure} = 0.1 \times (36)^{\frac{1}{2}} = 0.6$$

$$\text{Mass of } N_2(g) \text{ adsorbed per g of iron} = 0.6 \text{ g}$$

$$\text{Moles of } N_2(g) \text{ adsorbed per g of iron} = \frac{0.6}{28} = \frac{3}{140}$$



6.(D) Reactivity for nucleophilic addition reaction $\propto$ electrophilicity of carbon of carbonyl group.

7.(ACD) Volume of HCP unit cell = base area $\times$ height

$$= 6\sqrt{3}r^2 \times 4r\sqrt{\frac{2}{3}} = 24\sqrt{2}r^3$$

8.(AB) (C) For exothermic reaction activation energy may be smaller or greater than heat of reaction.
(D) when $T \rightarrow \infty$, $k = A$

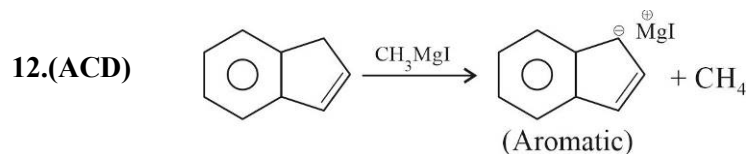
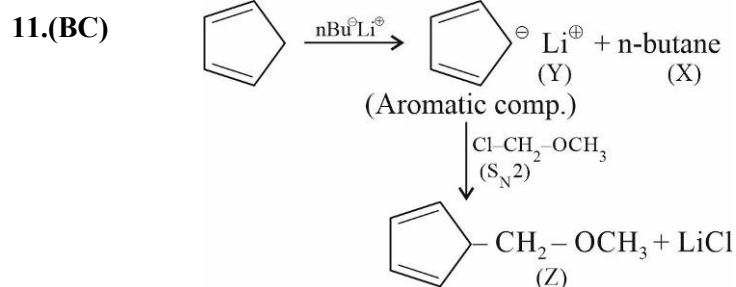
$$\mu = \frac{k_{T+10}}{k_T} = \frac{A}{A} = 1$$

9.(CD) (C) CMC value of $\text{CH}_3(\text{CH}_2)_9 \text{NH}_3^+ \text{Cl}^-$ will be higher than that of $\text{CH}_3(\text{CH}_2)_{12} \text{COO}^- \text{Na}^+$
(D) $i < 1$ in case of colloidal solution formation

10.(AC) $w_{\text{Zn}} = \frac{65.4}{2 \times 96500} \times 0.25 \times 9.65 \times 60 \times 60 = 2.943\text{g}$

$$w_{\text{MnO}_2} = \frac{87}{1 \times 96500} \times 0.25 \times 9.65 \times 60 \times 60 = 7.83\text{g}$$

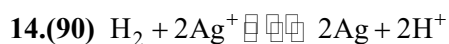
$$w_{\text{NH}_4^+} = \frac{18}{1 \times 96500} \times 0.25 \times 9.65 \times 60 \times 60 = 1.62\text{g}$$



13.(0.16) $\Delta T_f = K_f \cdot m$

$$26.84 - 25.64 = 8 \times \frac{(2.4 \times 10^{-3})}{M_o \times (100 \times 10^{-3})} \times 1000$$

$$M_o = 160\text{g/mol} = 0.16\text{kg/mol}$$



$$E_{\text{cell}} = 0.5 = (0.8 - 0) - \frac{0.06}{2} \log_{10} \frac{1}{1 \times [\text{Ag}^+]^2}$$

$$\Rightarrow [\text{Ag}^+] = 10^{-5} \text{ M}$$

$$\text{No. of moles of Ag in 1 lit. solution} = 10^{-5} \text{ mol}$$

$$\text{No. of moles of Ag in 250 ml solution} = \frac{10^{-5}}{4} \text{ mol}$$

$$w_{\text{Ag}} = \frac{10^{-5}}{4} \times 108 \text{ g} = 0.27 \text{ mg}$$

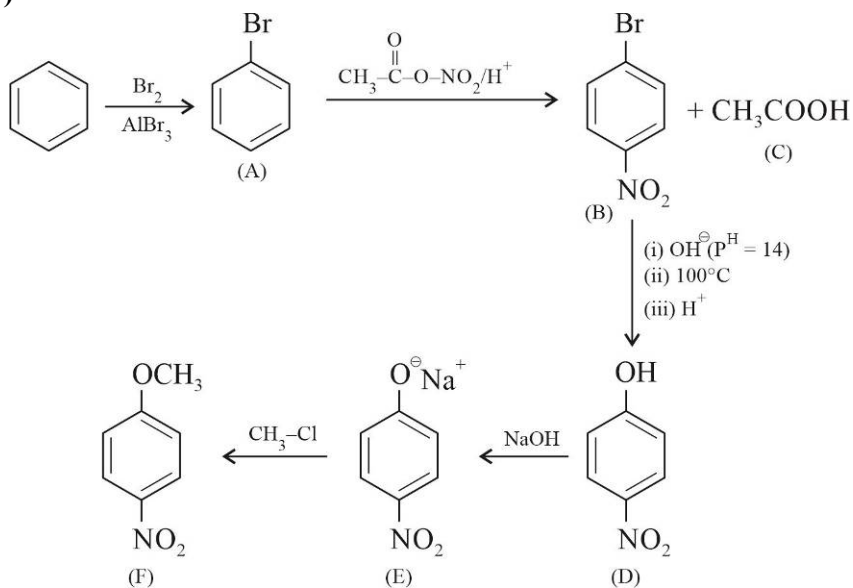
$$\% \text{ Pb in alloy} = \left(\frac{2.7 - 0.27}{2.7} \right) \times 100 = 90\%$$

15.(40) Distance between nearest neighbours $= 2r = \frac{a}{\sqrt{2}} = \frac{4\sqrt{2}}{\sqrt{2}} = 4 \text{ nm} = 40 \text{ \AA}$

16.(75) $\% \text{ of } S_N2 \text{ mechanism} = \frac{(6 \times 10^{-5})[\text{R}-\text{X}] \times 0.01}{\left[(6 \times 10^{-5})[\text{R}-\text{X}] \times 0.01 \right] + \left[(2 \times 10^{-7})[\text{R}-\text{X}] \right]} \times 100 = \frac{6}{8} \times 100 = 75\%$

17.(5) Refer S_NAr reaction.

18.(153)



MATHEMATICS

1.(D) $f'(x) = A \cdot \frac{\pi}{2} \cos\left(\frac{\pi x}{2}\right)$

$$f'\left(\frac{1}{2}\right) = A \frac{\pi}{2} \cos \frac{\pi}{4} = \frac{A\pi}{2\sqrt{2}} = \sqrt{2} \quad \therefore \quad A = \frac{4}{\pi}$$

$$\int_0^1 f(x) dx = \left[-\frac{2A}{\pi} \cos\left(\frac{\pi x}{2}\right) + Bx \right]_0^1$$

$$\frac{2A}{\pi} + B = \frac{2A}{\pi} \quad \therefore \quad B = 0$$

2.(C) $x|x| - x^2 |x| \geq 0 \Rightarrow |x|(x - x^2) \geq 0$

$$x = 0, 1 \text{ or } x - x^2 > 0 \Rightarrow 0 < x < 1$$

$$f(x) = \begin{cases} x|x| & , \quad 0 \leq x \leq 1 \\ x^2|x| & , \quad x < 0 \text{ or } x > 1 \end{cases}$$

$$g(x) = \begin{cases} x^2|x| & , \quad 0 \leq x \leq 1 \\ x|x| & , \quad x < 0 \text{ or } x > 1 \end{cases}$$

$$\int_{-1}^0 (x^2|x| - x|x|) dx + \int_0^1 (x|x| - x^2|x|) dx = \int_{-1}^0 (x^2 - x^3) dx + \int_0^1 (x^2 - x^3) dx = \int_{-1}^1 (x^2 - x^3) dx = 2 \int_0^1 x^2 dx = \frac{2}{3}$$

3.(B) Equation of circle will be $x^2 + (y-2)^2 + \lambda(y-2) = 0$

Differentiating, we get $2x + 2(y-2) \frac{dy}{dx} + \lambda \frac{dy}{dx} = 0$

$$\therefore \quad \text{the equation is } x^2 + (y-2)^2 - (y-2) \left(2x \frac{dx}{dy} + 2y - 4 \right) = 0$$

4.(A) Slope $= \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{y-1}{x^2+x}$

$$\Rightarrow \frac{dy}{y-1} = \frac{dx}{x^2+x} \Rightarrow \int \frac{1}{y-1} dy = \int \frac{1}{x} - \frac{1}{x+1} dx + C \Rightarrow \frac{(y-1)(x+1)}{x} = k$$

Putting $x=1, y=0$, we get $k=-2$ The equation is $(y-1)(x+1) + 2x = 0$

5.(A) $3\vec{i} + 2\vec{j} - 5\vec{k} = \lambda(2\vec{i} - \vec{j} + \vec{k}) + \mu(\vec{i} + 3 - 2\vec{k}) + \nu(-2\vec{i} + \vec{j} - 3\vec{k})$

$$3 = 2\lambda + \mu - 2\nu; 2 = -\lambda + 3\mu + \nu; -5 = \lambda - 2\mu - 3\nu$$

$$\Rightarrow \mu = 1; \nu = 2, \lambda = 3$$

$$\mu, \frac{\lambda}{2}, \nu \text{ are in A.P.}$$

6.(A) $(\vec{a} + \vec{b}) \times (\vec{a} \times \vec{b}) = \vec{a} \times (\vec{a} \times \vec{b}) + \vec{b} \times (\vec{a} \times \vec{b}) = (\vec{a} \cdot \vec{b})\vec{a} - |\vec{a}|^2 \vec{b} + |\vec{b}|^2 \vec{a} - (\vec{a} \cdot \vec{b})\vec{b}$

$$= \vec{a}\vec{b}(\vec{a} - \vec{b}) - \vec{b} + \vec{a} \quad (|\vec{a}| = |\vec{b}| = 1)$$

$$= ((\vec{a} \cdot \vec{b}) - 1)(\vec{a} - \vec{b})$$

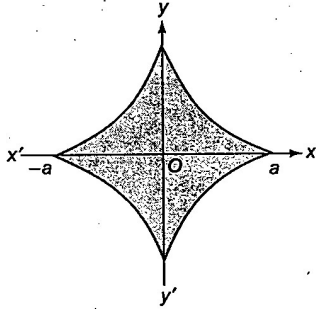
Hence $(\vec{a} + \vec{b}) \times (\vec{a} \times \vec{b})$ is parallel to $\vec{a} - \vec{b}$

$$7.(BC) \quad 1 = \frac{2}{1-e} \int \frac{dt}{t^2 + \left(\frac{1+e}{1-e}\right)} \quad (t = \tan x/2)$$

If $0 < e < 1$, $\frac{1+e}{1-e} > 0$, So, (B) is correct

If $e > 1$, $\frac{1+e}{1-e} < 0$, So, (C) is correct

$$8.(ACD) \quad \text{Eliminating } t, \text{ we have } x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}} \Rightarrow y = (a^{2/3} - x^{2/3})^{3/2}$$



From diagram,

$$\Rightarrow A = 2 \int_{-a}^a (a^{2/3} - x^{2/3})^{3/2} dx = 4 \int_0^a (a^{2/3} - x^{2/3})^{3/2} dx ; \quad A = 4 \int_0^a y dx = 4a^2 \int_0^{\pi/2} 3 \cos^3 t \sin^2 t \cos t dt$$

$$9.(ABC) \quad (A) \quad f(\lambda x, \lambda y) = \frac{\lambda(x-y)}{\lambda^2(x^2+y^2)} = \lambda^{-1} f(x, y) \Rightarrow \text{homogeneous of degree } (-1)$$

$$(B) \quad f(\lambda x, \lambda y) = (\lambda x)^{1/3} (\lambda y)^{-2/3} \tan^{-1} \frac{x}{y} \\ = \lambda^{-1/3} x^{1/3} y^{-2/3} \tan^{-1} \frac{x}{y} = \lambda^{-1/3} f(x, y) \Rightarrow \text{homogeneous}$$

$$(C) \quad f(\lambda x, \lambda y) = \lambda x \left(\ln \sqrt{\lambda^2(x^2+y^2)} - \ln \lambda y \right) + \lambda y e^{x/y} \\ = \lambda x \left[\ln \left(\frac{\lambda \sqrt{x^2+y^2}}{\lambda y} \right) \right] + \lambda y e^{x/y} = \lambda \left[x \left(\ln \sqrt{x^2+y^2} - \ln y \right) + y e^{x/y} \right] = \lambda f(x, y) \Rightarrow \text{homogeneous}$$

$$(D) \quad f(\lambda x, \lambda y) = \lambda x \left[\ln \frac{2\lambda^2 x^2 + \lambda^2 y^2}{\lambda x(x+y)} \right] + \lambda^2 x^2 \tan \frac{x+2y}{3x-y} \\ \lambda x \left[\ln \frac{2x^2 + y^2}{x(x+y)} \right] + \lambda^2 x^2 \tan \frac{x+2y}{3x-y} \Rightarrow \text{non homogeneous}$$

$$10.(AD) \quad \text{The given differential equation is } y_2(x^2+1) = 2xy_1 \Rightarrow \frac{y_2}{y_1} = \frac{2x}{x^2+1}$$

Integrating both sides, we get

$$\log y_1 = \log(x^2+1) + \log C$$

$$\Rightarrow y_1 = C(x^2+1) \quad \dots\dots\dots(1)$$

It is given that $y_1 = 3$ at $x = 0$

Putting $x = 0, y_1 = 3$ in equation (1), we get $3 = C$

Substituting the value of C in (1), we obtain

$$y_1 = 3(x^2 + 1) \quad \dots\dots\dots(2)$$

Integrating both sides w.r.t to x, we get

$$y = x^3 + 3x + C_2$$

This passes through the point (0, 1). Therefore, $1 = C_2$

Hence, the required equation of the curve is $y = x^3 + 3x + 1$

Obviously it is strictly increasing form equation (2)

Also $f(0) = 1 > 0$, then the only root is negative

11.(AB) $y = f(x)$ is a curve

$y - y = f'(x)(X - x)$ is tangent at (x, y)

$$f'(x) = \frac{dy}{dx} \quad X = 0$$

Initial ordinate of tangent $y - x f'(x)$

The subnormal at this point is given by $y \frac{dy}{dx}$; $y \frac{dy}{dx} = \frac{y}{x + y}$

This is homogeneous equation $\frac{dx}{dy} = \frac{x + y}{y} = \frac{x}{y} + 1$

$\frac{dx}{dy} - \frac{x}{y} = 1$ this is linear equation also

12.(AC) Let the required vector be $\vec{d} = x\hat{i} + y\hat{j} + z\hat{k}$, For this to be coplanar with $\vec{b}$ and $\vec{c}$, we must have

$$\begin{vmatrix} x & y & z \\ 1 & 2 & -1 \\ 1 & 1 & -2 \end{vmatrix} = 0$$

$$\Rightarrow x(-4 + 1) + y(-1 + 2) + z(1 - 2) = 0$$

$$\Rightarrow -3x + y - z = 0 \quad \dots\dots\dots(i)$$

The projection of $\vec{d}$ on $\vec{a}$ is $\frac{|\vec{a} \cdot \vec{d}|}{|\vec{a}|}$

$$\text{So } \sqrt{\frac{2}{3}} = \frac{1}{\sqrt{6}} |2x - y + z| \Rightarrow 2x - y + z = \pm 2 \quad \dots\dots\dots(2)$$

The choices (A) and (C) satisfy the equations (1) and (2)

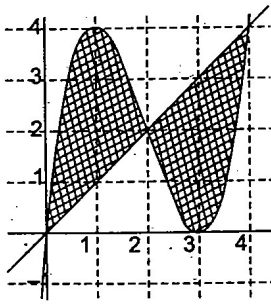
$$13.(-7) \quad f'(x) = 3x^2 \sin \frac{1}{x} - x \cos \frac{1}{x} \Rightarrow f(x) = \int \left(3x^2 \sin \frac{1}{x} - x \cos \frac{1}{x} \right) dx$$

$$f(x) = \sin \frac{1}{x} x^2 - \int \cos \frac{1}{x} \left(-\frac{1}{x^2} x^2 \right) dx - \int x \cos \frac{1}{x} dx$$

$$f(x) = x^3 \sin \frac{1}{x} + c$$

$$f(0) = 0 + c = 0 \Rightarrow c = 0 \quad \therefore f\left(\frac{1}{\pi}\right) = \frac{1}{\pi^2} \cdot \sin \pi + 0 = 0$$

14.(8) Required area $= 2 \int_0^2 (x(x-3)^2 - x) dx = 8$ sq. units



15.(1) We can write $y = A \cos(x+B) - Ce^x$

Where $A = c_1 + c_2, B = c_3$ and $C = C_4 e^{c_4}$

$$\frac{dy}{dx} = -A \sin(x+B) - Ce^x$$

$$\Rightarrow \frac{d^2 y}{dx^2} = -A \cos(x+B) - Ce^x \Rightarrow \frac{d^2 y}{dx^2} + y = -2Ce^x$$

$$\Rightarrow \frac{d^2 y}{dx^2} + \frac{dy}{dx} = -2Ce^x = \frac{d^2 y}{dx^2} + y \Rightarrow \frac{d^2 y}{dx^2} - \frac{d^2 y}{dx^2} + \frac{dy}{dx} - y = 0$$

Which is a differential equation of degree 1.

16.(1.60) Put $y = vx$

Equation become $\frac{dx}{x} = \left(\frac{1}{\sqrt{v^2-1}} - \frac{1}{v} \right) dv$ on integration $xy = y + \sqrt{y^2 - x^2}$

For $x=1, y=1$ we have $c=1$ $\therefore$ for $y=2$ We have $5x=8 \Rightarrow x = \frac{8}{5} = 1.60$

17.(0.75) $\vec{a} \times (\vec{a} \times \vec{c}) + 3\vec{b} = 0$

$$(\vec{a} \cdot \vec{c})\vec{a} - (\vec{a} \cdot \vec{a})\vec{c} + 3\vec{b} = 0$$

$$(2\sqrt{3} \cos \theta)\vec{a} - 3\vec{c} + 3\vec{b} = 0$$

$$(2 \cos \theta)\vec{a} - \sqrt{3}\vec{c} + \sqrt{3}\vec{b} = 0$$

$$|(2 \cos \theta)\vec{a} - \sqrt{3}\vec{c}|^2 = |1 - \sqrt{3}\vec{b}|^2$$

$$12 \cos^2 \theta - 249 \cos^2 \theta + 9 = 0 \quad \therefore \cos^2 \theta = \frac{9}{12} = \frac{3}{4}$$

18.(0.10) $\vec{\alpha} \cdot \vec{\beta} = \left(\frac{\hat{i}}{a} + \frac{4\hat{j}}{b} + b\hat{k} \right) \cdot \left(b\hat{i} + a\hat{j} + \frac{1}{b}\hat{k} \right) = \frac{b}{a} + \frac{4a}{b} + 1$

$$\frac{1}{5 + \alpha \cdot \beta} = \frac{1}{5 + \frac{b}{a} + \frac{4a}{b} + 1} = \frac{1}{\frac{b}{a} + \frac{4a}{b} + 6}$$

Then the minimum value of $\frac{b}{a} + \frac{4a}{b}$ is 4 $\therefore$ Maximum value of $\frac{1}{5 + \alpha \cdot \beta}$ is 0.10